

José María Barbeito

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EDUCATION

North Carolina State University, *Raleigh, NC*

- **B.S.E:** Chemical Engineering; **Minor:** Material Science Engineering *Expected May 2025*
- **Masters:** Material Science Engineering *Expected May 2026*
- **Current Cumulative GPA:** 3.7/4.0

INDEPENDENT PROJECT EXPERIENCE

Aquarium and Vivarium Construction, Fabrication Engineer, *Durham, NC*

March 2018 - Present

- Constructed a 55 gallon poison dart frog vivarium to house five distinct varieties of poison dart frogs, designed to be a self-sustaining closed cycle.
- Applied wet lab experience to create copper sulfate crystal structures at home.

Arduino Projects

August 2021- Present

- Built a self-controlling evaporator to dry fruit controlled by an Arduino, including a power switch, a heat bulb, and temperature and humidity sensors.
- Constructed various counters and clocks with LED screens and controlled by various mechanical inputs

Liquid Bromine Distillation

December 2021

- Distilled 0.98 moles of liquid bromine from sodium bromide using a variety of acids, reagents, and reactants. Utilized proper safety procedures and PPE.

WORK EXPERIENCE

Process Engineering Intern, Wolfspeed, *Durham, NC*

May 2023 – Present

- Utilized statistical process control and experimental design to fine tune machines and eliminate issues in the SiC wafering process, including software analysis, DOE testing methods, and mechanical trials.
- Worked closely with engineers with various projects with regards to specific processes required in the making of silicon carbide wafers.
- Currently designing and testing new proprietary system to enhance yield.

Doble W Servicios, *Buenos Aires, Argentina*

February 2021 – March 2021

- Disassembled and reassembled various cars and trucks to install bulletproof glass and kevlar plating for the Argentine Police - averaging ten cars a day fully outfitted.
- Constructed bulletproof glass through assembling and kilning multiple layers of glass and specialty polycarbonate before sanding and installing.

LEADERSHIP AND COMMUNITY EXPERIENCE

TSDCH Chemistry Club, Leader and Co-founder, *Durham, NC*

August 2019 – April 2020

- Founded and led the Chemistry Club at TSDCH.
- Organized a program for first graders to learn about phase changes through super-saturated solutions and gas-producing reactions.

BEE Educated, Founder, *Durham, NC*

August 2014 – Dec 2020

- Founder and lead instructor of BEE Educated, a local outreach program that teaches children ages 8-12 the importance of bees through an on-site program with active apiaries.
- Helped local beekeepers with their own hives, teaching and assisting them in keeping healthy colonies, along with providing swarm capture services for locals.

TECHNICAL SKILLS: Woodworking, Metalworking, Lathe, Crystalline Structure Fabrication, Drone/Digital Cinematography

SOFTWARE SKILLS: JMP, SPACE, Proprietary manufacturing systems, Microsoft Office Tools, JavaScript and JSL

LANGUAGES: Spanish (Native), German (Beginner)

INTERESTS: Computer Assembly, Backpacking, Traveling, Rock Climbing, Scuba Diving, Drums, Guitar, Crystal Growing